

CONTACT INFORMATION	Department of Mathematics The Ohio State University 231 W. 18TH Ave Columbus, OH 43210 United States	Email: luo.1409@osu.edu Phone: +1 (614) 769-8800 Webpage: https://luowynorman.github.io https://math.osu.edu/people/luo.1409
RESEARCH INTEREST	Optimal transport, dimension-free optimization, symmetry, manifold learning, high-dimensional probability.	
EDUCATION	The Ohio State University, Columbus, Ohio, United States Ph.D. in Mathematics (Applied Track) Thesis Advisor: Dustin G. Mixon	2022 - now
	The Chinese University of Hong Kong, Shenzhen, China B.Sc. in Mathematics and Applied Mathematics (First Class Honours)	2018 - 2022
PUBLICATIONS	1. On the clustering behavior of sliding windows (with Boris Alexeev, Dustin G. Mixon, Yan X. Zhang) <i>SIAM Journal on Mathematics of Data Science</i> , 2026, [Accepted Manuscript][arXiv]	
PREPRINTS	1. BalLOT: Balanced k -means clustering with optimal transport (with Dustin G. Mixon) Revised manuscript under review at <i>Mathematical Programming</i> , 2026, [Latest Draft][arXiv] 2. Optimal transport bounds for autoencoder-based density smoothing on MTW manifolds (with Wonjun Lee) Manuscript submitted to <i>Journal of Machine Learning Research</i> , 2026, [Latest Draft][arXiv]	
AWARDS & FELLOWSHIP	Special Graduate Assignment The Ohio State University, Department of Mathematics Undergraduate Research Award The Chinese University of Hong Kong, Shenzhen Academic Performance Award (Top 5%) The Chinese University of Hong Kong, Shenzhen	Spring 2027 2021 2020 - 2021
ACADEMIC ACTIVITIES	Summer School in High-Dimensional Probability and Its Applications Northwestern University, Evanston SLMath Workshop on Statistical Optimal Transport University of California, Berkeley MREG Workshop on Multiclass Perceptron University of Michigan, Ann Arbor	August 2026 June 2025 August 2023
INTERDISCIPLINARY EXPERIENCE	Multi-omics integration platform for prognostic biomarker discovery [GitHub Link] Data Science Bootcamp, Erdős Institute	
		November 2025

TEACHING	The Ohio State University		
	Recitation Instructor		
	Math 1148 College Algebra (Instructor: Amit Vutha)	Spring 2026, Autumn 2026	
	Math 1156 Calculus for Biology (Instructor: Trent Ohl)	Autumn 2023	
	Math 2153 Calculus III (Instructor: Amit Vutha)	Spring 2023	
	Math 1151 Calculus I (Instructor: Prerona Dutta)	Autumn 2022	
	Grader		
	Math 3345 Foundations of Higher Mathematics (Instructor: Weichen Gu)	Autumn 2025	
	Math 2568H Honors Linear Algebra (Instructor: Dustin Mixon)	Autumn 2025	
	Math 5522H Honors Complex Analysis (Instructor: Jennifer Park)	Spring 2025	
	Math 5530H Honors Probability (Instructor: Neil Falkner)	Spring 2025	
	Math 6211 Real Analysis I, PhD-Level (Instructor: David Penneys)	Spring 2024	
	Math 2568 Linear Algebra (Instructor: Dustin Mixon)	Spring 2024	
	The Chinese University of Hong Kong, Shenzhen		
	Undergraduate Student Teaching Fellow		
Real Analysis (Instructor: Wei-Ming Ni)	Spring 2021		
MENTORING PROGRAM	The Ohio State University		
	Directed Reading Program		
	Probability, Concentrations, and Programming		
	Mentee: Avery Stevens	Fall 2023	
PROGRAMMING LANGUAGES	Python, MATLAB, R, LaTeX		
LANGUAGES	English (fluent), Chinese (native in Cantonese and Mandarin)		